

Datamining

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Introduction

ESS ERIC is an international research initiative whose main work is the ‘European Social Survey’ (see [European Social Survey European Research Infrastructure \(ESS ERIC\) 2023](#)) whose main objective is to capture changing opinions and values in European countries. The eighth edition of the survey which is the data-set that we’ll be using is built on observations collected from 2016 to 2017. It spans across 23 countries and the scope of the population is all people aged 15 or more, part of a private household.

Methodologically, the authors of the study opted for a random sampling of participants, with a minimum of 70% positive responses to the interview request, in order to minimize as much as possible the risks of self-selection bias that could render the conclusions drawn from the study misinformed. The survey includes a weight variable to adjust observations significance in representing society as a whole.

One of the main themes of the study is attitudes towards social assistance in the context of a changing Europe. Respondents are asked to indicate whether they are in favor of various specific welfare policies which are described to them. The question we will seek to answer in this paper is : Can we predict what an individual’s opinion of basic income will be using data-mining techniques ?

Context

Universal income is a social assistance policy that consists of distributing a basic income to all members of a community, with no eligibility restrictions and therefore no targeting.

Universal income is not a new idea - it has been around since at least the 18th century - but it is a policy proposal that is now at the height of its popularity. It’s a idea that appeals to a number of political camps for different reasons: it has a much smaller bureaucratic weight than other programs, it reduces inequalities by being a massive redistributive policy, and it provides insurance against the automation of work. It would therefore be interesting to see

which political opinions and other survey variables are associated with a favorable opinion of universal income implementation.

The following is the definition of basic income that the authors of the survey use for participants who speak English : “The government pays everyone a monthly income to cover essential living costs. - It replaces many other social benefits. - The purpose is to guarantee everyone a minimum standard of living. - Everyone receives the same amount regardless of whether or not they are working. - People also keep the money they earn from work or other sources. - This scheme is paid for by taxes.” (see [European Social Survey European Research Infrastructure \(ESS ERIC\) 2020](#))

Methodology

The aim of our application of data-mining techniques is to conduct predictive modeling to classify, as accurately as possible, the type of the observation from the value of its explanatory variables. In this case the type we seek to predict is whether the individual is favorable to the institution of a universal basic income. In a previous work we examined what factors could affect the probability of an observation being favorable through the use of a logistic regression. Despite finding useful results, the logistic regression cannot offer us very tangible insights on how accurate the model is on new data-points. This is a shortcoming that limits our ability to apply the model to project how policy support would change if demographics change. Another advantage of data-mining techniques is the ability to use more explanatory variables, both quantitative and qualitative, where in the previous work we had to heavily modify the original data-set to make the data appropriate for a logistic regression.

Despite the robustness of techniques like classification trees when faced with a great quantity of variables, we still have to remove the vast majority from the data-set considering there are 535 different variables, there would be no case where a reasonably sized and not over-fitted model would need them all. The essential variables that we’ve selected are the level of education, gender, age, living with a partner, having a paid job, and political spectrum affiliation.

The modifications we brought to the variables seek to harmonize the various responses, some qualitative variables are not completely ordinal for example, like the response variable : ‘basic’ that has 4 levels of support from ‘Strongly against’ to ‘Strongly in favour’ but from values 5 to 7 of the variable the neutral or lack of response is represented. Considering the object of our interrogation is to discover how to best identify a supporter of the policy, and to have a framework to examine the consequence of changing demographics, we decided to transform it into a binary variable representing support for the policy. We think the theoretical implications are coherent, factors such as path dependency make us think that for a policy to be implemented it needs a strong base of supporters and indifference is practically similar to disapproval.

Models

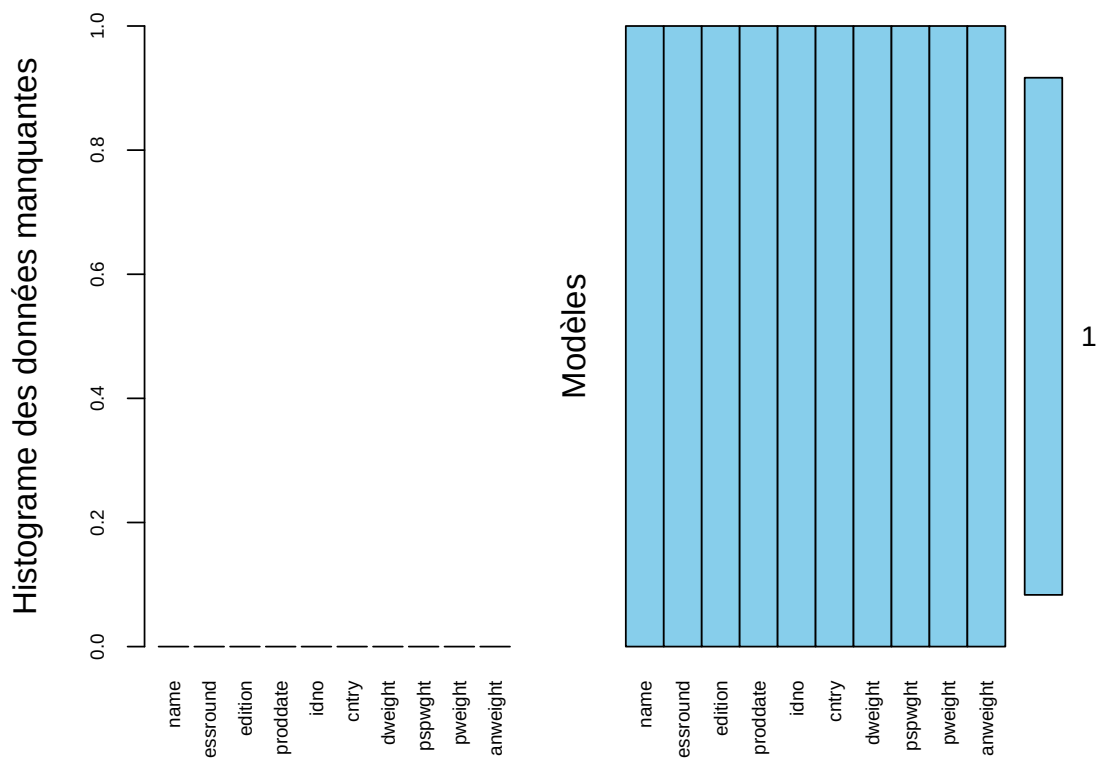
Descriptive statistics

1. Distribution of basinc

basinc	anweight	percentage
1	5158	9.92
2	15135	29.11
3	22166	42.63
4	4189	8.06
7	293	0.56
8	5048	9.71
9	5	0.01

The majority of respondents (50.63%) said they were in favor or strongly in favor of implementing a universal income. However, a significant proportion (39.04%) expressed opposition, with 9.92% being strongly opposed. In addition, 10.33% of responses included missing or neutral values, which could reflect indecision or a lack of understanding of the concept.

2 Missing values



Variables sorted by number of missings:

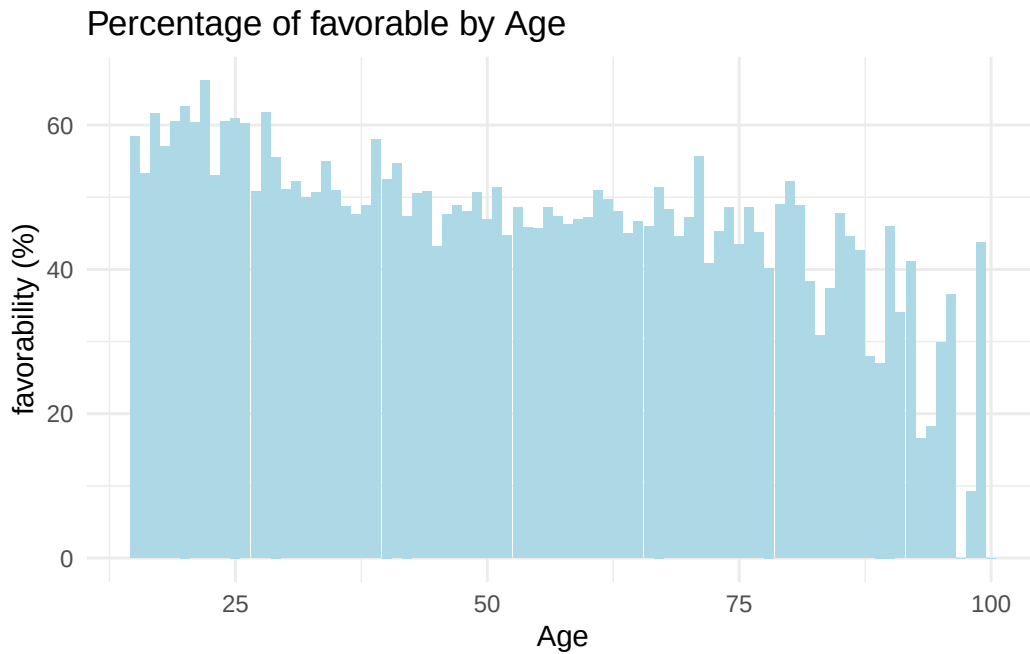
Variable	Count
name	0
essround	0
edition	0
proddate	0
idno	0
cntry	0
dweight	0
pspwght	0
pweight	0
anweight	0

This chart shows there are no missing values for any of the variables we've chosen. This is

because when there are missing values in this data-set they are handled with special values that vary for each variable.

3. Relations between basic income and other variables

3.1 By age



Women show a slight tendency towards higher support, with 52.4% and 51.6% of responses in the “In favor” and “Strongly in favor” categories, respectively. Conversely, men are slightly over-represented in the “Strongly against” category (51.7%). These results suggest a slightly more favorable attitude to universal income among women.

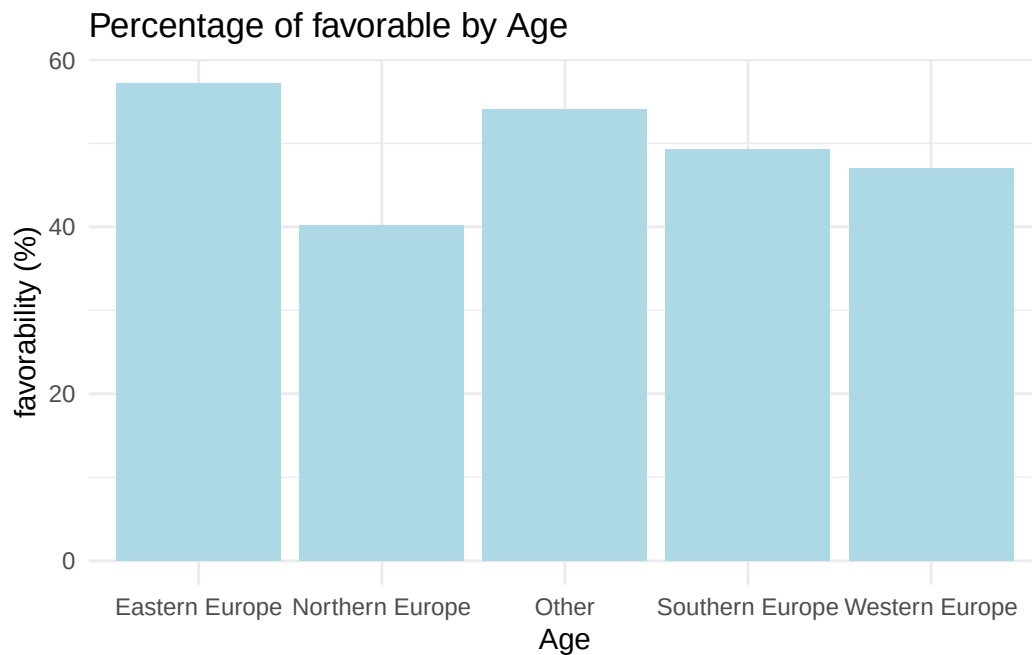
3.2 By education

eisced	anweight	percentage
1	5310	10.21
2	10264	19.74
3	7177	13.80
4	13031	25.06
5	6111	11.75
6	3368	6.48
7	6485	12.47

55	122	0.24
77	71	0.14
88	51	0.10
99	6	0.01

Cross-analysis of baseline with education reveals a strong influence of education level on the acceptability of universal income. Respondents with higher levels of education (6, 7) show a greater propensity to support the idea, particularly in the “In favor” and “Strongly in favor” categories (3 and 4). Conversely, opposition is more marked among intermediate and lower levels (3, 4, 5). These results suggest that higher levels of education may be linked to a better understanding or more positive acceptance of redistributive policies such as universal income.

4. Analysis by country



Central and Eastern European countries, such as the Czech Republic and Russia, stand out for their stronger opposition, while countries like Israel and Lithuania show strong support in the ‘In favor’ and ‘Strongly in favor’ categories. The Nordic countries, notably Finland and Norway, show high levels of moderate support, while Iceland is less committed overall. These regional trends highlight the impact of cultural, economic and political contexts on the acceptability of universal income.

Classification models

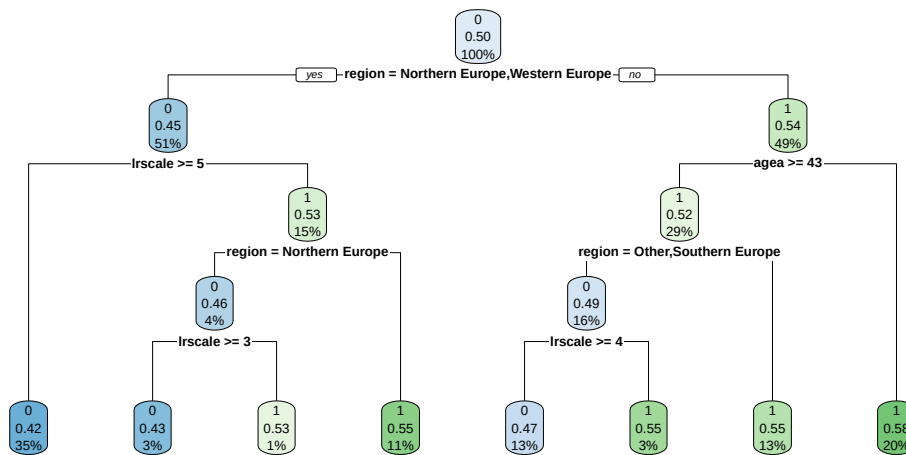
In this section, we complete the descriptive analysis by using decision trees to explain opinion on universal income. This method is consistent with our research question, since we are seeking to identify which factors (level of education, gender, age, etc.) influence opinion on basic income. In addition to being non-parametric and easy to interpret, the decision tree makes it possible to segment the population into groups with different probabilities of supporting basic income.

```
data_dm <- report

# Construction de l'arbre sur l'ensemble complet
tree_full <- rpart(basinc ~ eisced + gndr + agea + icpart1 + pdwrk + lrscale + region,
  data = data_dm,
  method = "class",
  control = rpart.control(maxdepth = 4,
    cp = 0.001))

# Affichage de l'arbre
rpart.plot(tree_full, main = "Decision tree")
```

Decision tree



Results

Our objective is to have a rate of correctly classified individuals over 50%, a model that performs better than pure chance. Through the use of decision trees, we've managed to achieve that and determine which factors can best identify an observation's class.

Country of origin plays a key role in the perception of universal income. It seems the easiest way to classify an individual's probability of supporting universal income is by looking at which region he is from.

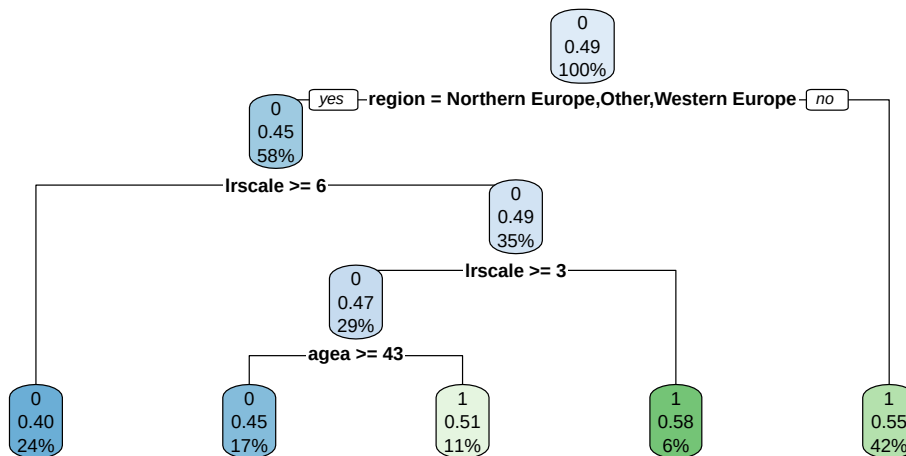
Secondly, political orientation 'lrscale' strongly influences results: individuals on the left are generally more favorable, while those on the right are more reticent.

Global error rate : 43.67 %

Training set size : 22116 10

Validation set size : 22116 10

Arbre de décision (Training Set)



Error rate on training sample : 44.11 %

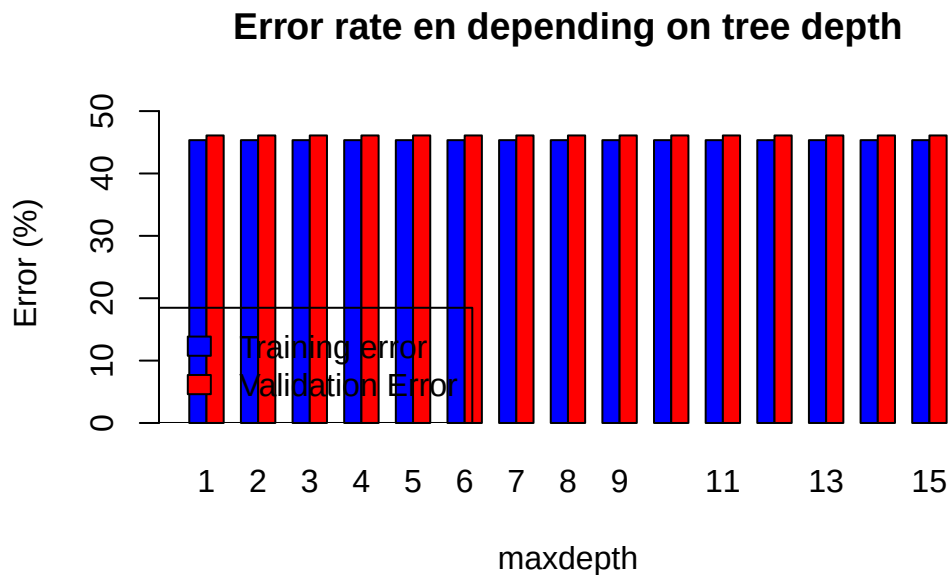
Error rate on validation sample : 44.39 %

Error rate for basinc = 0 on training : 52.96 %

Error rate for basinc = 1 on training : 35.03 %

Error rate for basinc = 0 on validation : 53.29 %

Error rate for basinc = 1 on validation : 35.44 %



We can see here that the error rates remain relatively constant, suggesting that the maxdepth modification has had no significant impact on model performance. The lack of variation in the curves may indicate a stable model, but also that there has been no real optimization by increasing maxdepth parameter. The decision tree model is consistent and shows no overfitting, but its accuracy remains limited.

Conclusion

Considering the results of the various trees we've built, we see there is little possibility for highly accurate classification using the current technique and set of data. Our aim, to have a model that can determine with less than 50% error has been achieved, with the global error at 44% and the error rate for detecting observations with favorable opinions on basic income much lower than for those who don't.

Maybe the understanding of the influence of different factors could be improved by removing regions from the available variables, as regional differences might capture many differences caused by other variables.

For policy it seems that changes in the political landscape of Europe might be the greatest factor in determining if basic income has a future as a policy proposal, as it is the most important variable, that has the ability to change.

European Social Survey European Research Infrastructure (ESS ERIC). 2020. “ESS8 Data Documentation.” Sikt - Norwegian Agency for Shared Services in Education; Research. <https://doi.org/10.21338/NSD-ESS8-2016>.

———. 2023. “ESS8 - Integrated File, Data Set.” Sikt - Norwegian Agency for Shared Services in Education; Research. https://doi.org/10.21338/ess8e02_3.